

control of semiconductor qubits using superconducting resonators.

What the future holds

With all the work going on in the field of quantum computing, the tech landscape will witness a major shift within the decade.

The emergence of fault-tolerant logical qubits: Within the next 5-10 years, the focus will shift from improving physical qubits to constructing the first ‘logical qubits’. A logical qubit is an abstraction formed from many entangled physical qubits, which are collectively used to encode and protect a single piece of quantum information from errors, enabling truly fault-tolerant quantum computation.

Distributed quantum computing via a ‘quantum internet’: The development of reliable quantum interconnects will lead to the first small-scale quantum networks. These networks will connect multiple quantum processors located in different labs or data centres, enabling distributed quantum algorithms and laying the groundwork for a future ‘quantum internet’ with applications in secure communication and sensing.

AI-driven discovery of quantum materials and architectures: AI will become an indispensable tool for designing the next generation of quantum hardware. Machine learning models will be used to sift through vast parameter spaces to discover novel materials with optimal properties for qubits and to design entirely new qubit architectures that are inherently more robust against noise.

‘Quantum-ready’ semiconductor foundries: Major semiconductor foundries will establish dedicated fabrication lines for quantum integrated circuits. This will commoditise the production of quantum chips, making powerful quantum hardware more accessible and affordable, and accelerating the pace of innovation across the entire quantum ecosystem.

The rise of topological qubits:

While still in the research phase in 2025, the long-term goal is the realisation of topological qubits. These qubits encode information in the collective, global properties of a system, making them naturally immune to local sources of noise. If successful, topological quantum computing would offer a more direct path to fault tolerance, potentially leapfrogging other qubit modalities.

Qubits are the building blocks of quantum computers, and several competing technologies are currently in play. Superconducting circuits, which use tiny loops of superconducting material, are among the most prominent. Companies like IBM and Google have developed powerful processors using this approach. Trapped ions present another pathway, where individual atoms are suspended using electromagnetic fields and manipulated with lasers. This technique offers high fidelity and stability. Photonic qubits, which use particles of light, promise scalability and room-temperature operation, with firms like Xanadu and PsiQuantum championing this technology. Other notable approaches include spin qubits in semiconductors and topological qubits, each bringing unique strengths and challenges to the table.

One of the greatest challenges in quantum hardware is maintaining the delicate quantum states that enable computation. Quantum noise and decoherence—the process by which quantum information is lost to the environment—can quickly disrupt calculations. Researchers employ error correction techniques, advanced shielding, and refined control systems to preserve coherence. Innovations in materials science and circuit design have made strides, but the battle against noise remains an ongoing challenge. As quantum computers scale up, managing these effects becomes even more vital.

To keep quantum hardware stable, especially superconducting qubits,

scientists often rely on cryogenics—ultra-cold environments that can reach temperatures close to absolute zero. Special refrigerators, known as dilution refrigerators, create these conditions, reducing thermal noise and allowing quantum effects to flourish. The design of quantum environments extends beyond cold temperatures, incorporating vacuum chambers and electromagnetic shielding to protect against outside disturbances. These specialised setups are crucial for reliable and repeatable quantum operations.

Building a practical quantum computer means going beyond a handful of qubits. Modular architecture, where several smaller quantum processors are linked together, represents a promising approach to scaling up. Integration with classical electronics, improved interconnects, and advances in quantum networking enable larger and more powerful systems. Researchers are also exploring error-tolerant designs and hybrid platforms that blend different qubit technologies for greater flexibility.

Companies and research labs at the forefront of quantum computing

The quantum hardware race is driven by a mix of established firms, ambitious startups, and world-class research institutions. IBM, Google, and Intel are among the giants developing superconducting and silicon-based quantum processors. IonQ and Honeywell lead the field in trapped ion technology, while Rigetti Computing pursues modular quantum architectures. European labs, including those at Delft University and the University of Oxford, have made significant contributions. In Asia, institutions like the Chinese Academy of Sciences and Japanese R&D centres are accelerating progress. Collaboration between academia, industry, and government is spurring innovation worldwide.

Looking ahead, quantum hardware is poised for rapid evolution.